Tenacious Conversion Engine - Act V Decision Memo [DRAFT]

Executive Summary

Build: signal-grounded outbound pipeline; measure reply rate, stalled-thread rate, and CPL.
Result: tau2 pass@1 0.450 vs baseline 0.400 (Delta A=+0.050).
Recommendation: 30-day Segment 2 pilot, 60 leads/week, $8.00/week cap, kill-switch at >= 5% wrong-signal.

1) Benchmark Performance

- Method vs baseline (sealed tau2): 0.450 vs 0.400 (Delta A=+0.050).
- Delta A significance (Fisher one-sided): p=0.500 (sig<0.05? False).
- 95% CIs: method [0.250, 0.650]; baseline [0.200, 0.600].
- Auto-opt baseline: 0.400. Published reference: 0.42.
- tau2 runtime (not email latency): p50=27.46s, p95=61.53s.

2) Cost per Qualified Lead (CPL)

- Qualified lead: booking_created == true (non-autoresponder inbound replies).
- Cost inputs: LLM=$0.23; rig=$3.48; APIs=$3.30. Total=$7.01.
- Denominator: 22 qualified leads (booked=22).
- Derived CPL: $7.01 / 22 = $0.32 - within target (<= $5.00).
- Target envelope (Tenacious internal, baseline_numbers.md): <= $5.00; penalty >= $8.00.

Caching note: LLM routes via OpenRouter -> qwen/qwen3-235b-a22b; prefix caching unverified for this route - LLM cost is an upper bound. Reduction options: shorter prompts, smaller model, deterministic rules for simple classifications.

3) Stalled-Thread Rate

- Definition: stalled = booking_created is false (14-day no-booking proxy).
- Measured: 80.5% (91/113 non-booking; booked=22).
- Tenacious manual baseline (given): 30.0%-40.0%.

Caveat: synthetic traces + booking proxy; production calibration required.

4) Reply-Rate Delta (signal-grounded vs generic)

- Competitive-gap variant: 100.0% (27/27)
- Generic variant: 14.0% (30/215)
- Delta (cg - generic): 86.0% pp. Top-quartile benchmark: 7.0%-12.0%.

Caveat: 100% reply rates are a synthetic artifact; treat delta as directional only.

5) Annualized Revenue Impact

- Segment 2: 2 calls/wk -> $1,440,000-$10,800,000/yr
- Segment 1, Segment 2: 6 calls/wk -> $4,320,000-$32,400,000/yr
- Segment 1, Segment 2, Segment 3, Segment 4: 12 calls/wk -> $2,880,000-$64,800,000/yr

6) Pilot Recommendation (30 days)

- Segment: Segment 2. Justification: cost-pressure buyers; fastest CFO sign-off.
- Volume: 60 leads/week. Budget cap: $8.00/week.
- Success criterion: >=5 discovery calls booked with qualified Segment 2 prospects within 30 days

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A) Four Failure Modes tau2 Does Not Capture

1) Wrong-signal outreach: weak signal -> confident claim -> brand + reply-rate collapse.
   Fix: confidence-tier phrasing + abstain threshold. Cost: more checks, lower volume.
2) Tone offense: offshore/spam language triggers hiring managers.
   Fix: style rubric + pre-send review.
3) Proxy mismatch: booking_created != qualified in Tenacious CRM.
   Fix: explicit stage mapping before pilot launch.
4) Bad benchmark fit: top-quartile practice wrong for specific sub-niche.
   Fix: calibrate benchmark to niche before prescribing gap.

B) Public-Signal Lossiness

- False negative: silent company -> underestimated readiness -> weak pitch -> lost deal.
- False positive: loud company -> overestimated readiness -> technical pitch -> brand hit.
- FN agent shape: pitches generic, misses Segment 4 gap -> wasted touch + lower replies.
- FP agent shape: pitches advanced gap to low-readiness prospect -> credibility + spam hit.

C) Gap-Analysis Risks (when top-quartile is a bad benchmark)

- Risk 1: deliberate strategy (e.g. regulated workflow) -> move-fast rec harms trust.
- Risk 2: sub-niche irrelevance (gap not the binding constraint) -> message flagged spam.

D) Brand-Reputation Unit Economics

- Assume 1,000 sends; 5% wrong-signal rate = 50 wrong-signal emails.
- Expected brand cost at $10k pipeline suppression/wrong-signal email = $500k.
- Compare to reply-rate lift: 7-12% (signal-grounded) vs 1-3% (generic baseline).

E) Unresolved Probe - signal_overclaiming (P-005)

- Trigger rate: 9/9 in probe suite. Impact: undermines ’grounded research’ value prop.
- If deployed at scale: can revert reply rates to baseline + create viral brand risk.
- Unit economics: 5% wrong-signal on 1,000 sends -> ~$500k expected brand cost (see D).

F) Kill-Switch Clause

- Metric: wrong-signal rate on manually audited first 100 sends.
- Threshold: pause if wrong-signal >= 5% OR any high-severity exec complaint received.
- Rollback: revert to generic template + require human approval for signal-grounded claims.